

# Fiber mirror photothermal effects on optical spring

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## Abstract

Optical spring effect occurs when circulating light in a cavity (formed by two mirrors facing each other with trapped light inside) exerts a cavity length-dependent radiation pressure force on a movable mirror. These systems have demonstrated their capability for highly sensitive measurements in gravitational wave detection. However, they face stability challenges due to the cavity's finite response time. We aim to study photothermal response of fiber mirror coatings, where the thermorefractive effect has previously been shown to contribute to spring stability. In this paper, we designed a testing instrument for the photothermal effect. The instrument fixes fibers and aligns them on a v-groove using a Teflon plate and applies pressure with screws. This ensures quick switching and alignment of fiber mirrors. With this system, we created a cavity of length  $28(1)\mu m$  and finesse  $3.51(7) \times 10^4$ . The high finesse ensures a sensitive response of cavity power with respect to detuning. Furthermore, we successfully implemented laser locking at room temperature and pressure, achieving a feedback bandwidth of 3.3 kHz, with the phase shift  $\phi_G = -\pi$  occurring at 70(1) kHz. This configuration suppresses thermal noise within the control bandwidth and provides a stable platform for subsequent photothermal effect measurements.

# 1 Introduction

The "optical spring" system exists in various forms; Fabry–pérot cavity is one example, where two mirrors face each other to create an optical cavity, with one mirror attached to the mechanical resonator. When the mirror oscillates, it modulates the cavity's circulating power, which in turn alters the radiation pressure force acting on the mirror, creating an optical spring effect.

Optical springs have numerous applications. They have been utilized to enhance the sensitivity of gravitational wave detectors at LIGO laboratory[1]. Recent research also demonstrates their ability to enable programmable defect modes in mechanical crystals[2]. This technique can be further used for storing and transporting quantum information.

Due to the finite response time of light within the cavity to the mirror's motion, a lag is introduced in the optical spring response. This delay causes a destabilizing "antidamping" effect. Currently, this challenge has been addressed by injecting a phase delay in the laser frequency-locking servo, which can enhance damping at the cost of optical rigidity[3] – resulting in greater displacement of the optical spring at a given force. Alternative methods include implementing active feedback loops to counteract the lagging response[4]. However, these loops require real-time measurement of the cavity's length and corresponding adjustments to the laser power or cavity length.

Photothermal response of the fiber mirror coating can influence the cavity length, which consequently affects the circulating power. The photothermal effect comprises two components: thermoelastic deformation, which shortens the cavity, and the thermorefractive effect, which changes the refractive index of the coating and effectively lengthens the cavity. Previous observations have shown that when the thermorefractive effect dominates, it contributes to a stable optical spring.

In this paper, we develop an instrument that enables quick switching and alignment of fiber mirrors, facilitating efficient testing and potential use in similar applications. A traditional method of aligning fiber mirrors uses two ferrules mounted in a C-shaped sleeve, where fibers are glued in the ferrules [5], making it difficult to switch to other fiber mirrors.

In Section 2.1, we introduce our instrument design and design considerations. In Section 2.2, we present our optical diagram and the concept of the feedback loop used for laser-locking. In Section 2.3, we describe a method for measuring the cavity length and its finesse. In Section 3.1, we present results of the finesse measurement. In Section 3.2, we display a power spectral density (PSD) plot of our lock-in system. In Section 3.3, we measure transfer function of our system.

## 2 Materials and Methods

### 2.1 Switchable fiber mirror alignment instrument

The designed instrument has three parts - the top view of the base block is shown in Figure 1a), and the side view with two top bars attached to the base block is shown in Figure 1b). The separate design makes it easier to glue components and align fibers. The experiment was originally intended to be tested in a vacuum chamber to minimize thermal noise. Therefore, all screw holes are tapped all the way through to avoid trapping air in the block.

The base block has a roughly cubic shape, which enhances stability by distributing mechanical stress more evenly. Bottom four screw holes are used to mount the base on an optical breadboard. The middle slit is designed for inserting a membrane chip. Two shear piezos are glued onto the base block, they can move toward or away from the slit when voltage is applied.

The v-groove was originally one piece; we drilled it from the middle after gluing it onto the piezos to ensure proper alignment.

The two top bars can be connected to the base block using screws. These bars are designed with equal width and height to increase stability. The screws pass through the top bar holes and apply pressure on the Teflon cover to secure the fibers in the v-groove. This ensures that the fibers move with the v-groove (driven by the piezos) rather than sliding along it. This screw-based fixing method also allows for easy replacement of the fiber mirror in the system. The actual picture of the instrument is shown in Appendix A.

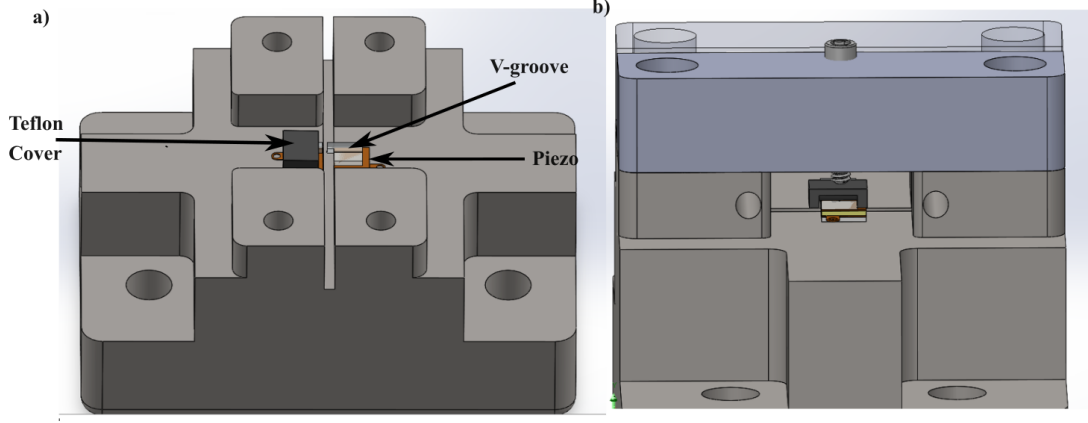


Figure 1: The design of our vacuum-compatible instrument enables quick switching and alignment of fiber mirrors. Figure a) shows the top view of the base block. Fibers are aligned using a v-groove. Beneath the v-groove is a piezo actuator which can move the v-groove, thereby adjusting the fiber position to change the cavity length. A Teflon cover placed over the v-groove, with Teflon being a vacuum-compatible material. There are 4 tap holes on the bottom of the block, which allow the block to be fix on optical breadboard. Figure b) the side view of the instrument. The middle screw from the top bar is used for applying pressure on the Teflon cover, securing the fibers to the v-groove. This ensures that the fibers move with the v-groove rather than sliding along it when the piezo actuator is in operation.

## 2.2 Optical diagram and feedback loop

We use 1550nm laser in this experiment, with a phase modulator to modulate the laser frequency. The optical diagram is shown in Figure 2, along with a feedback loop.

We need to lock the cavity at its resonance peak to perform further experiments. Only when the cavity is on resonance does it provide a sensitive response between the cavity length and cavity power. Even when the cavity is initially on resonance, detuning noise (from thermal fluctuations and laser frequency variations) can push the cavity off resonance. We use a feedback loop to correct the cavity length and suppressed detuning noises.

As shown in Figure 2b, detuning ( $\delta$ ) enters the cavity ( $C$ ), changes the cavity power, and is converted to an electronic signal by the photodiode ( $D$ ). This signal is amplified and filtered by the servo controller ( $-A$ ), then sent to the piezo ( $P$ ) to change the cavity length and adjust detuning ( $\delta$ ). The loop equation yields  $\delta = \delta_n - CDAP\delta$ , where  $C, D, -A, P$  are complex function of frequency. Solving for  $\delta$  gives:

$$\delta = \frac{\delta_n}{1 + CDAP}. \quad (1)$$

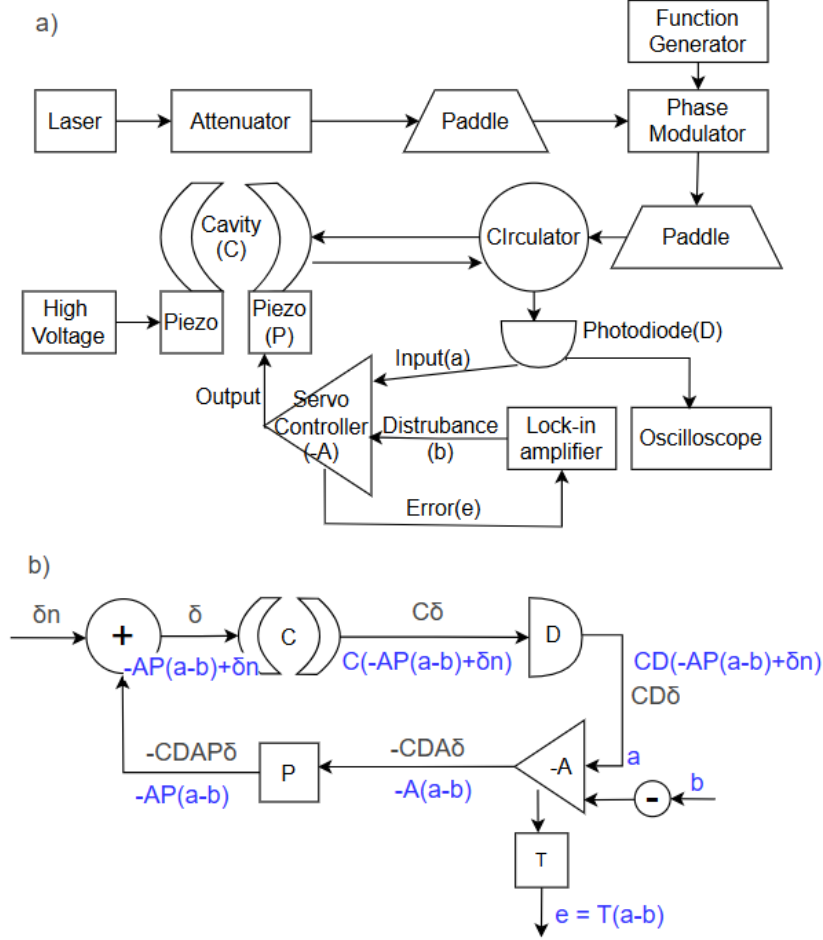


Figure 2: Figure a) shows our optical diagram: The power of the 1550nm laser is modulated by an attenuator. A paddles change the polarization of the laser to match the response characteristics of the following phase modulator. The phase modulator is controlled by a sine wave from a function generator. After the phase modulator, the polarization is adjusted again to correspond to cavity polarization before passing through a circulator and into the cavity. The reflected power from the cavity is detected by a photodiode and sent to both an oscilloscope and a servo controller. Output from the servo controller is fed to the piezo to modulate the cavity length and correct for noise. A 200V high-voltage supply is connected to another piezo. This is used for changing the cavity length and locating the resonance peak. Figure b) shows the feedback loop: the black feedback path shows the feedback loop for correcting detuning noise ( $\delta_n$ ). Noise ( $\delta_n$ ) enters the system and is converted to an optical signal by the cavity (transfer function  $C$ ). This signal is detected by a photodiode ( $D$ ) and manipulated by the servo controller ( $-A$ ), then sent back to the piezo ( $P$ ) to correct the noise input, yielding loop equation:  $\delta = \delta_n - CDAP\delta$ . The feedback path in blue shows the measurement of transfer function ( $CDAP$ ) with injected disturbance ( $b$ ). The cavity power signal from the photodiode ( $a$ ) connects to the positive input of the servo controller, while the disturbance ( $b$ ) connects to the inverted input of the servo controller. The error monitor port of the signal is sent to a lock-in amplifier with readout  $e = T(ab)$ , where  $T$  is the transfer function of the lock-in amplifier. The servo controller amplifies the input ( $-A(a-b)$ ) and sends the output to the piezo, which responds ( $-AP(a-b)$ ). Detuning noise ( $\delta_n$ ) together with the piezo response ( $-AP(a-b)$ ) +  $\delta_n$  enters the cavity ( $C$ ) and modulates the cavity power ( $C((-AP(a-b)) + \delta_n)$ ), then gets read out from the photodiode ( $a = CD((-AP(a-b)) + \delta_n)$ ), which is sent back to the servo controller.

We want to make close-loop gain  $G \equiv CDAP$  as large as possible to maximally cancel the noise. However, time delays in the signal path induce phase shifts ( $\phi_G$ ). When  $\phi_G = -\pi$  at

a finite frequency, the denominator in Eq. 1 becomes negative, and we need to ensure  $G < 1$  when  $\phi_G = -\pi$  to avoid instability of the system. Since most noise occurs at low frequencies, the ideal feedback loop should have  $\phi_G = -\pi$  occurring at high frequencies, with gain ( $G$ ) rapidly decreasing as it approaches frequency  $\phi_G = -\pi$ .

To measure the closed-loop transfer function ( $CDAP$ ), we use a lock-in amplifier. A lock-in amplifier is an instrument used to extract signals buried in noise. With a given reference signal, usually the same signal used to modulate or drive the system, the lock-in amplifier can extract the amplitude and phase of the output signal that corresponds to the reference signal frequency.

We use the lock-in amplifier (Zurich Instruments HF2LI) to inject a known disturbance ( $b$ ) into the system, which serves as the reference signal. The reflected power from the cavity ( $a = \delta CD$ ) is connected to the positive input of the servo controller, while the disturbance ( $b$ ) is connected to the inverted input. The error monitor output from the servo controller is  $(a - b)$ , which is passed back to the lock-in amplifier to get readout  $e = T(a - b)$ , where  $T$  is the transfer function of the lock-in amplifier. The output of the servo controller  $u = -A(a - b)$  is fed back to the piezo (P). The new loop equation is  $e = T((-AP(a - b) + \delta_n)CD - b)$ , solving for  $e$  yields:

$$e = \frac{-T}{1 + APCD}b + \frac{TCD}{1 + APCD}\delta_n. \quad (2)$$

The noise  $\delta_n$  term is uncorrelated with the disturbance ( $b$ ) and can be eliminated when the lock-in amplifier extracts the signal. The circuit transfer function can therefore be expressed as:

$$APCD = -\frac{Tb}{e} - 1. \quad (3)$$

The only unknown is  $T$ , the transfer function of the lock-in amplifier. This can be measured independently by sending the disturbance ( $b$ ) directly back into the lock-in amplifier, resulting in readout  $e = Tb$ , where  $T = \frac{e}{b}$ .

## 2.3 Finesse measurement

To characterize the cavity we set up with our instrument, we measure its finesse. Finesse is a dimensionless quantity that equals  $2\pi$  times the average number of round trips photons travel inside the cavity before they escape or decay. Higher finesse means photons bounce many times, leading to sharper resonance. Finesse  $\mathcal{F}$  can be calculated using [6]:

$$\mathcal{F} = \frac{\pi c}{L\kappa}, \quad (4)$$

where  $c$  is the speed of light,  $L$  is the length of the cavity, and  $\kappa$  is the power decay rate, which equals the full width at half maximum (FWHM) of the resonance peak.

To determine the cavity length, we use image analysis. Knowing the diameter of the fiber is  $125\mu m$ , we can calculate the length of the cavity by comparing the pixel measurements of the fiber diameter and the cavity length.

To measure the FWHM of the cavity, we need to convert the signal from the photodiode, which is in the time domain, to the frequency domain. We achieve this by injecting sidebands to the cavity using a phase modulator. The laser electric field in the cavity is  $E = A \cos(w_l t)$ , where  $A$  is the amplitude of the electric field and  $w_l$  is the laser frequency. By sending a

modulation frequency  $w_m$  to the phase modulator, we modulate the phase of the electric field, yielding:

$$E = A \cos(w_l t + B \sin(w_m t)), \quad (5)$$

$$E \approx A \cos(w_l t) - \frac{AB}{2} \cos((w_l - w_m)t) + \frac{AB}{2} \cos((w_l + w_m)t). \quad (6)$$

Eq. 6 is obtained using trigonometric identities and a small-angle approximation for the  $B \sin(w_m t)$  term. As a result, in addition to the resonance peak corresponding to the laser frequency ( $w_l$ ), the cavity will also have resonance peaks at frequencies  $w_l \pm w_m$ .

The reflected power at resonance  $P(r)$  is given by [6]:

$$P(r) = \frac{1 + \frac{2a}{\kappa} \Delta}{1 + \frac{4}{\kappa^2} \Delta^2}, \quad (7)$$

where  $\Delta$  is the detuning with respect to the resonance, and  $a$  is the asymmetric parameter for 3D resonance peak. To fit Eq. 7 to our time-domain data with phase-modulated sidebands at  $w_l \pm w_m$ , we modify it to:

$$A \frac{1 + \frac{2a}{w}(t - t_0)}{1 + \frac{4}{w^2}(t - t_0)^2} + B \frac{1 + \frac{2a}{w}(t - t_0 + d)}{1 + \frac{4}{w^2}(t - t_0 + d)^2} + B \frac{1 + \frac{2a}{w}(t - t_0 - d)}{1 + \frac{4}{w^2}(t - t_0 - d)^2}, \quad (8)$$

where  $A$  and  $B$  are the magnitudes of the resonance peaks,  $t_0$  is the time position of the central resonance peak,  $w$  is FWHM of the resonance in time and  $d$  is the time shift of the phase-modulated resonance peaks.

By fitting Eq. 8 to the resonance peaks in our data, we can determine both the FWHM and the time difference ( $d$ ) between adjacent resonance peaks. Knowing the phase-modulation frequency ( $w_m$ ), we can then calculate  $\kappa$  using:

$$\kappa = \frac{w_m}{d} \cdot w, \quad (9)$$

where phase-modulation frequency ( $w_m$ ) correspond  $d$  time shift in our data, then  $w \cdot w_m/d$  corresponding the decay rate  $\kappa$ .

## 3 Results

### 3.1 Calculating cavity finesse

We measure the pixel dimensions of the fiber diameter and cavity length ten times and obtain a fiber diameter of 71(1)px and a cavity length of 15.7(5)px, where the uncertainties are from the standard deviation of the measurements. Knowing the diameter of the fiber is  $125\mu m$ , we calculate the length of the cavity as  $\frac{125}{71(1)} \times 15.7(5) = 28(1)\mu m$ . Measurements and images of the cavity are shown in Appendix B.

We take one measurement with a 1GHz frequency injected into the phase modulator and two measurements with 500MHz. We then fit the signal from the oscilloscope with Eq. 8 to obtain the time difference ( $d$ ) and FWHM of the central resonance peak. An example of the measurement is shown in Figure 3.

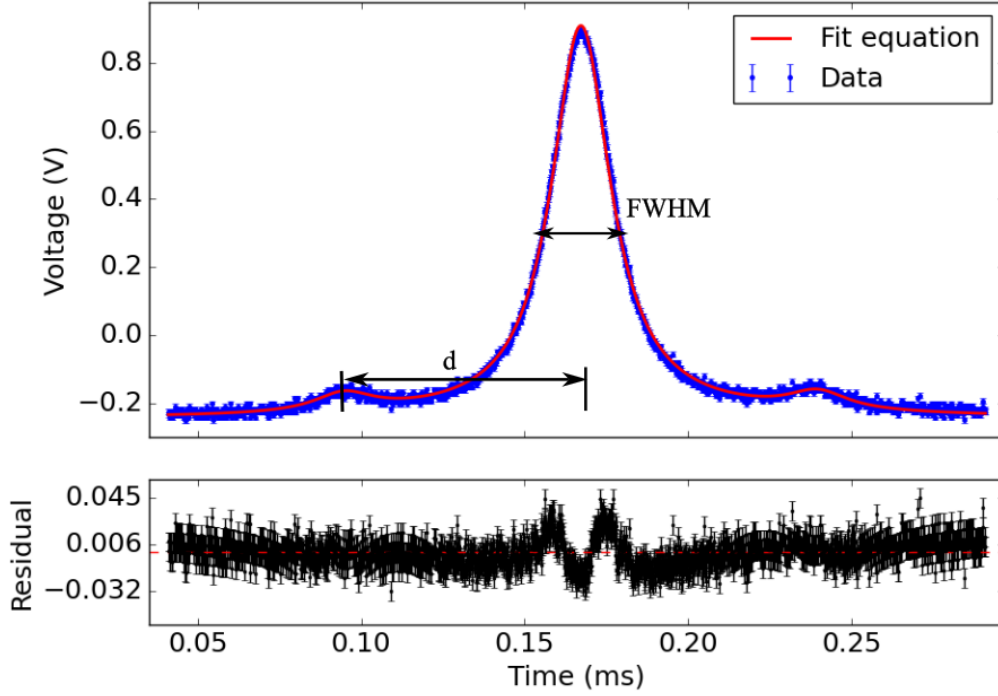


Figure 3: Central resonance peak at laser frequency ( $\omega_l$ ), and sidebands at frequencies ( $\omega_l \pm \omega_m$ ), where  $\omega_m = 500\text{MHz} \times 2\pi$ . The data is fitted with Eq. 8 to get FWHM and  $d$ , which are labeled in the figure. The error bars are calculated from the standard deviation of the data points when the cavity is off-resonance. The fit yields  $w = 2.211(3)\text{s}$  and  $d = 7.25(1)\text{s}$ .  $\chi_r^2 = 1.89(3)$ , indicating a reasonable fit.

After obtaining  $d$  and FWHM from the fit, we use Eq. 9 to calculate the values of  $\kappa$  are  $9.66(2) \times 10^8 \text{ rad/s}$ ,  $9.52(2) \times 10^8 \text{ rad/s}$ , and  $9.59(2) \times 10^8 \text{ rad/s}$ , with uncertainties derived from the fitting results. Using Eq. 4, the mean finesse of these three measurements is  $3.51(7) \times 10^4$ .

### 3.2 Power spectral density of lock-in

We use power spectral density (PSD) to characterize the noise frequency and locking performance. We first close the laser and take ten measurements of electronic noise (photodiode, oscilloscope) with sampling interval ( $\Delta t$ )  $1\mu\text{s}$  and 2500 sample points. This gives a frequency range of  $1/(2\Delta t) = 500 \text{ kHz}$  and frequency resolution of  $1/(N\Delta t) = 400 \text{ Hz}$ . Then, we turn on the laser and take ten measurements of laser noise; the average voltage for the laser power from the photodiode is  $1.363(3)\text{V}$ . We lock the cavity at the resonance peak where the power read from the photodiode is  $0.76(3)\text{V}$ .

Assuming the laser noise is classical noise, we can rescale the laser noise proportional to its power. The normalized PSD of electronic noise, rescaled laser noise, lock-in, and lock-in with electronic and laser noise removed is shown in Figure 4.

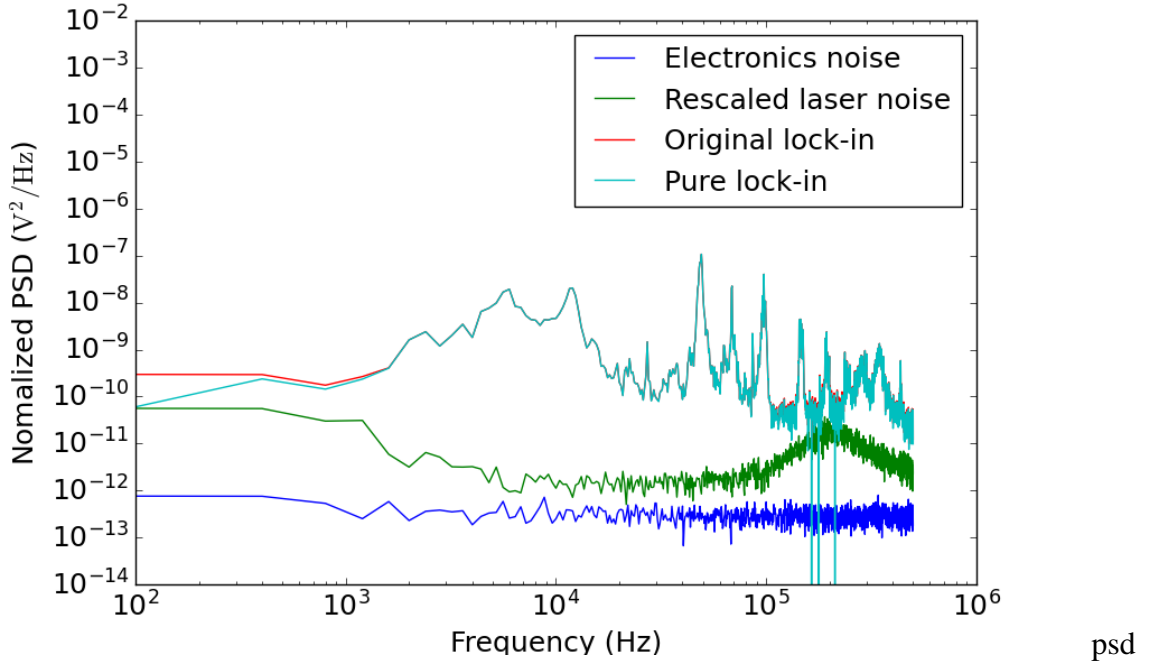


Figure 4: PSD of electronic noise, rescaled laser noise, original lock-in, and pure lock-in with electronic noise and laser noise removed. In the pure lock-in PSD, there are peaks occurring at 49kHz, 68kHz, 97(2)kHz, 146(4)kHz, 193(4)kHz, 243(5)kHz, 286(2)kHz, 345(2)kHz, and 435(1)kHz. The laser noise has a peak around 193(5)kHz, where the rescaled laser noise is higher than the lock-in noise, causing the plot to go down toward negative infinity on the log-log scale.

### 3.3 Transfer function

We first send a 100mV disturbance ( $b$ ) from the lock-in amplifier directly back into its input port to get a readout ( $e$ ), allowing us to calculate the transfer function  $T = e/b$ .

Then we send a 100mV disturbance ( $b$ ) to the inverted input of the servo controller and send the error signal ( $a - b$ ) from the servo controller back to the lock-in amplifier to get readout ( $e$ ). Using Eq. 3, we compute the closed-loop transfer function (CDAP). The results are shown in Figure 5.



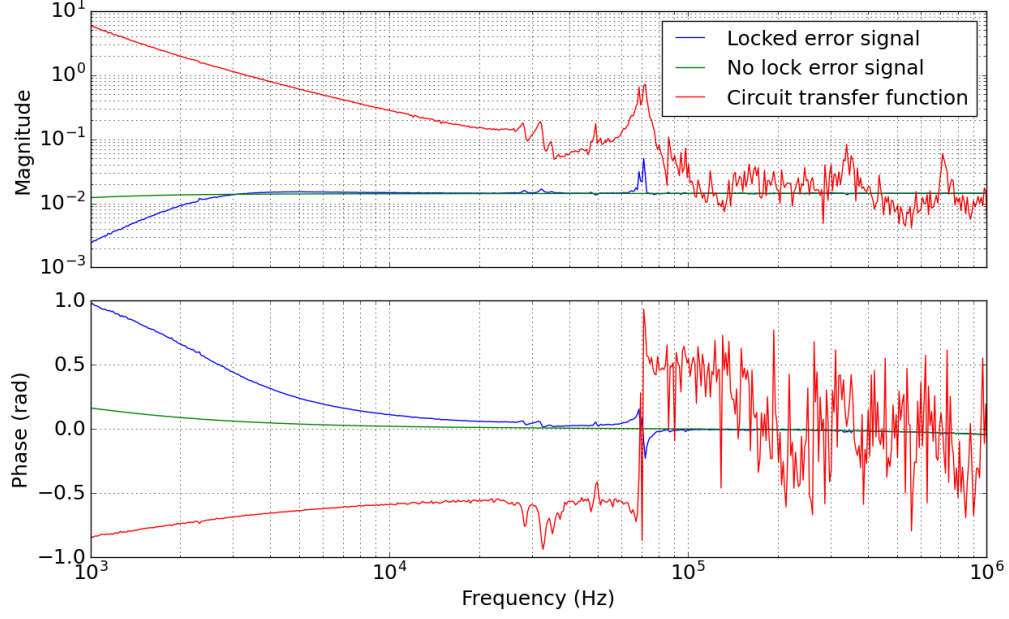


Figure 5: Closed-loop transfer function calculated from Eq. 3. "No lock error signal" is obtained by directly sending disturbance ( $b$ ) back into the lock-in amplifier. "Locked error signal" is obtained by sending disturbance ( $b$ ) to the inverted input port of the servo controller and feeding the error signal ( $a - b$ ) back into the lock-in amplifier. The phase shift reaches  $-\pi$  when the frequency reaches 70(1)kHz, corresponding to the ringing frequency. The bandwidth of the transfer function is at 3.3kHz. Before 3.3kHz, the magnitude of the "Locked error signal" is smaller than then "No lock error signal", so the transfer function effectively suppressed the error signal. It decreases the noise by about a factor of 10 at 1kHz, then about a factor of 100 at about 100Hz.

## 4 Discussion

High finesse means the cavity exhibits a more sensitive response to changes in cavity length, which enables higher sensitivity for detection. However, high finesse also makes it more difficult to lock the cavity on resonance. Successfully locking a high-finesse cavity thus demonstrates the quality of our design. It provides a stable system capable of achieving high-finesse laser locking at room temperature and pressure. This setup will be further used to test photothermal effects on different fiber mirror coatings.

The (PSD) in Figure 4 of the laser lock-in signal shows multiple peaks. These peaks may arise from mechanical resonances in our system. Possible sources include the piezo actuator, which exhibits a harmonic-oscillator-like transfer function, as well as mechanical modes of the moving fiber[7]. To identify which mechanical motions correspond to specific PSD peaks, we can simulate the mechanical resonances of the system to determine their frequencies. Additionally, by varying the clamping length of fibers and observing how the peak frequencies shift, we can determine whether they are associated with the fiber's mechanical modes.

From Figure 5, we observe that the phase of the transfer function reaches  $-\pi$  at a frequency of 70(1) kHz. This is the frequency at which ringing occurs. When  $\phi_G = -\pi$ , a positive gain  $G$  becomes negative, amplifying noise and potentially leading to instability and uncontrolled oscillations in the system. At 70(1) kHz, the magnitude of the gain is just below 1, which prevents ringing from occurring. Moreover, the direct resonance frequency, which corresponds to the highest peak of the system response, is around 70 kHz. This is not ideal, and we can resolve this by inserting a low-pass filter into the feedback circuit. This would cause the phase

$\phi_G$  to reach  $-\pi$  at a frequency lower than the resonance frequency. Avoiding the coincidence of direct resonance and the critical phase shift, and reducing the risk of instability.

The bandwidth of the transfer function is 3.3 kHz, defined as the frequency at which the gain magnitude drops to 1. This bandwidth determines the range of noise frequencies the feedback loop can effectively correct. For noise components below 3.3 kHz, the feedback loop suppresses and corrects the disturbances. For frequencies above 3.3 kHz, the correction is no longer effective. In future photothermal measurements of the fiber mirror coating, we will use a phase modulator to vary the laser frequency and measure the corresponding transfer function. To ensure that the injected laser frequency modulation is not corrected by the feedback loop, the modulation frequency must be above 3.3 kHz.

## 5 Conclusions

In this research project, we successfully designed an instrument that enables quick switching and alignment of fiber mirrors, which can be used for further testing of photothermal effects on different mirror coatings. The cavity length was measured to be  $28(1)\mu m$  with a finesse of  $3.51(7) \times 10^4$ . These measurements were obtained through image analysis and by using a phase modulator to create sidebands in the cavity. The high finesse ensures sensitive measurements with respect to changes in cavity length. In addition, we successfully locked our cavity on the resonance peak with a 3.3 kHz bandwidth and phase shift  $\phi_G = -\pi$  at 70(1) kHz.

Future prospects for this project include using a phase modulator to modulate the laser frequency and measure the transfer function of the photothermal effect. By testing different mirror coatings and systematically etching the end mirror to vary the thickness of the final layer, we aim to enhance the thermorefractive effect, which contributes to a stable optical spring.

## A Actual instrument picture

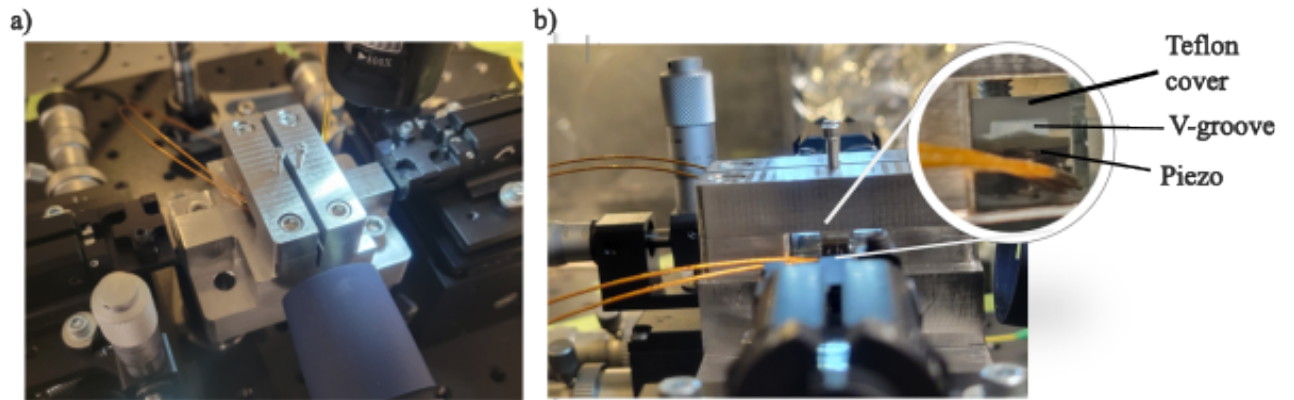


Figure 6: Figure a): the front and upper face of the stage. Two nanomax stage on two sides of the stage to help align fibers. Figure b): a side view of the block. A piezo is under a v-groove. Screws apply pressure on a teflon cover to clamp fiber in side the v-groove.

## B Image analysis of cavity length

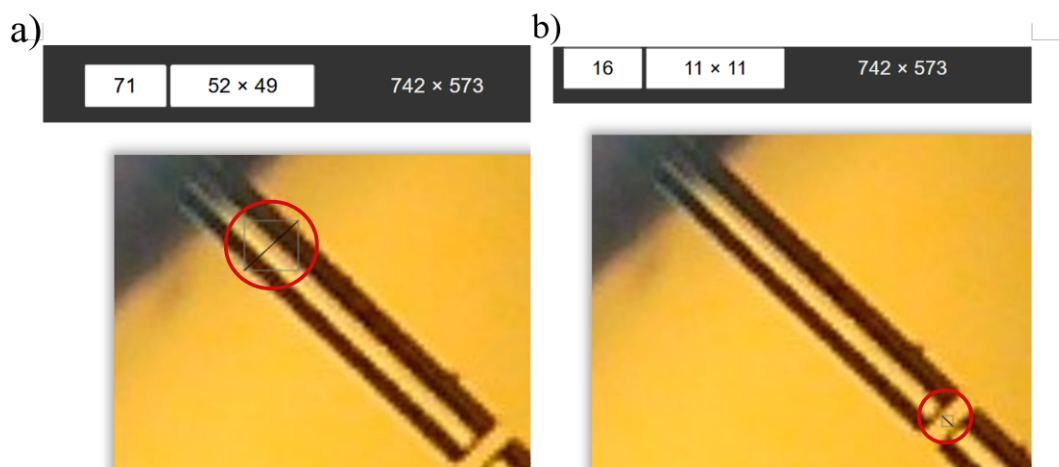


Figure 7: An example of using image analysis to measure the length of the cavity. Figure a): measuring the pixel of the diameter of the fiber, get 71px. Figure b): measuring the pixel of the length of the cavity, get 16px.

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